Journal of Archaeological Method and Theory (2026) 33:36
https://doi.org/10.1007/s10816-026-09771-7

RESEARCH

Sodium Chronicles... Grain by Grain. New Methods for Quantifying Ancient Salt Demand

A. Sotgia[1] · L. Alessandri[2] · P. A. J. Attema[1] · F. Bulian[1] · J. Sevink[3]

Received: 13 January 2026 / Accepted: 22 February 2026
© The Author(s) 2026

Abstract
Salt played a crucial role in ancient societies, serving both dietary and industrial purposes. Despite its importance, direct quantification of salt demand remains challenging due to limited archaeological evidence and the complexity of variables involved in salt production. This study proposes an indirect approach to estimating salt consumption through a reconstructive model of agropastoral landscape exploitation. By assessing the maximum number of people and animals requiring salt, along with the production of food items such as bread, cheese, and preserved meat, this model allows for a broad estimation of salt needs in each settlement. By reconstructing the territorial system's agricultural and pastoral economy, this research estimates salt consumption and explores its implications for settlement sustainability. The case study of the Iron Age site of 'Piscina Torta' (Rome, Italy)—a production site—is used to test this methodology. The study further assesses the necessary volume of seawater to meet the estimated demand. While the results are estimates, they provide a valuable reference framework for understanding salt consumption trends in early state development. Ultimately, this research highlights the significance of salt as a strategic resource, contributing to broader discussions on ancient economic systems and resource management.

Keywords Salt Demand Quantification · Ancient Salt Production · Agropastoral Landscape Modelling · Strategic Resource Control · Early State Political Economy · Iron Age Italy

✉ A. Sotgia
a.sotgia@rug.nl

[1] Groningen Institute of Archaeology, RUG - University of Groningen, Groningen, The Netherlands

[2] Department of Science of Antiquities, Sapienza University of Rome, Rome, Italy

[3] Institute for Biodiversity and Ecosystem Dynamics (IBED), University of Amsterdam, Amsterdam, The Netherlands

Published online: 05 March 2026

Introduction

Why Salt?

The centrality of sodium chloride (NaCl) in both human and animal diets, as well as its pivotal role in the major industries of the ancient world, has long been acknowledged (see Di Fraia (2011) and the 'Methodology: Model Description and Computational Approach' section). It is no coincidence that Homer, to emphasise its value within ancient societies, described this substance as 'divine' (*Iliad* 9.214), a linguistic marker that underscores the socio-cultural significance attributed to its consumption and control.

For the sake of practicality, the term 'salt' will be here used to refer specifically to sodium chloride, although the term in its broader chemical sense also encompasses solid compounds derived from acids and bases that dissociate into cations and anions where dissolution occurs.

In this text we will attempt to quantify salt importance especially in terms of actual consumption. We do so in the context of the rise of city-states in central Tyrrhenian Italy during protohistory, where control of this resource must have been of considerable importance in socio-political dynamics.

On the other hand, as argued by Carusi (2018:482), for Greek and Roman authors (*i.e.*, Aristotle, *Mirabilia* 138; Sallust, *Bellum Iugurthinum* 89.7–8; Varro, *De re rustica* I.7.8; Tacitus, *Annales* XIII.57; Appian, *Hispanica* 54.227; Synesius of Cyrene, *Epistulae* 148.4), the use of salt was considered one of the defining features of civilised life, distinguishing it from the practices of barbarian or foreign peoples who neither knew of it nor employed it. This literary *tópos* further illustrates the importance attached to salt in antiquity and confirms its legitimate place within studies of the economic exploitation of the ancient landscape. As highlighted in the recent book *Meridians of Salt: Global Perspectives on Archaeology and Ethnoarchaeology* (Eubanks *et al.*, 2025), salt production contributed—through processes that appear strikingly similar across the globe—to the establishment and expansion of economic and social exchange networks, to the development of new policies and economic practices, as well as intensifying competition and, in some cases, even violence among human communities.

Within this broader framework, the 'Salt and Power' project (Alessandri & Attema, 2021) seeks to reconstruct the ways in which control over salt production played a decisive role in the emergence and development of socio-political dynamics in Italy during the first half of the first millennium BCE. In the specific study area under consideration, Tyrrhenian central Italy, several settlement phenomena may indeed be interpreted through a 'salt explanation'. The expansion of settlement along the coasts from the tenth century BCE onwards (Belardelli *et al.*, 2021; Carusi, 2008a; Cinquegrana, 2025; Mandolesi, 2024; Pacciarelli, 1991), as well as the conflict between Rome and Veii for control over the Tiber River mouth (Dionysius of Halicarnassus, *Roman Antiquities* 2.55.5; Livy, *Ab Urbe Condita* 1.33), appears to be directly connected to the desire of nascent early states to secure dominance not only over other landscape resources but also, crucially,

over salt production. This argument holds true for other areas and chronological periods as well.

To adequately interrogate the role of salt in such transformative processes, purely qualitative approaches prove insufficient. A shift towards quantification is required, one that allows for the systematic measurement of demand thereby providing a more precise calibration of salt's socio-economic weight. Quantification, in this sense, is not a mere technical supplement to qualitative inquiry but a necessary analytical lens through which the magnitude of production, ownership, and exchange can be rendered empirically intelligible.

The pioneering attempt to theorise such a quantitative framework was undertaken by Moinier (2011) in *Salt in Antiquity: A Quantification Essay*, which constitutes a seminal reference point, and of which the present text may be regarded as both an update and a development. Building upon the premises set forth by Moinier (2011), the present work introduces an alternative method for estimating the salt requirements of a community.

Unlike the French scholar, who relied primarily on written sources to determine the variables necessary for calculating demand (*i.e.*, number of people or quantity of wheat produced), the methodology proposed here is grounded in the analysis of the landscape exploited by the communities under investigation. By reconstructing agropastoral production, understood as the quantity of cultivated crops, the number of animals raised, and the human population that could be supported by these resources, it becomes possible to obtain an indirect quantification of salt demand, using some of the same variables as Moinier did. In other words, the overall amount of salt required for the sustenance of both humans and livestock, as well as the quantities employed in specific productive activities, is estimated under the assumption that the community under analysis could meet such demand through its own productive system.

Numerous studies already provide estimates relating both to salt production in general and to specific productive applications ("Case Study: 'Piscina Torta'" section), and these serve as a valuable starting point for the elaboration of the model presented here ('Discussion' section). Once salt demand is established, it becomes possible to estimate the volume of seawater that would have been needed to be processed to obtain the required amount of salt, thereby further characterising the magnitude of salt production and its role within historical phenomena.

The application of this model in the simulation conducted ('Conclusion' section) for the territorial system encompassing the production site of 'Piscina Torta' (Alessandri *et al.*, 2024) demonstrates the heuristic efficacy of our quantitative methodology. The choice to focus on this case study is not accidental but is linked to the features investigated through multiple excavation campaigns by the University of Groningen (2022–2024). The discovery of a 15-hectare large site with numerous kilns and productive areas makes 'Piscina Torta' one of the best-known Iron Age salt-production workshops along the Tyrrhenian coast. However, more than a case-specific exercise, the simulation we present illustrates the methodological potential of quantification as a means of resituating salt within the material and political economies of the ancient Mediterranean.

Indeed, according to historical sources (Dionysius of Halicarnassus, *Roman Antiquities* 2.55.5; Livy, *Ab Urbe Condita* 1.33), the area where the productive site of 'Piscina Torta' was located initially (eighth–seventh century BCE) belonged to the small city of Ficana and its territorial system, before coming under the control of the emerging urban Rome, following the expansion of its territory towards the coast (seventh–sixth century BCE). These two scenarios were not selected simply to show, also quantitatively, the obvious correlation between demographic growth and the increase in salt demand recorded between the Iron Age and the Archaic period. Rather, they aim to highlight how the transition from medium-small communities to increasingly complex urban systems can also be read through the lens of changing scales of need (in this specific case, that of salt) and, consequently, of the productive strategies required to meet them, whether strictly technical (*i.e.*, changes in modes of production) or socio-political (*i.e.*, territorial expansion and conquest).

In other words, the quantification of salt demand and of the hypothetical production required to satisfy it in these two scenarios provides an additional tool for describing the impact that Rome's growth had within the broader processes of socio-economic transformation in Archaic Tyrrhenian Italy, as discussed in the concluding section of this study.

Uses of Salt

The multiplicity of uses of salt in antiquity has been extensively documented and may be briefly summarised as follows:

Salt must first and foremost be understood as a vital nutritional resource for humans. Its physiological functions are fundamental: it regulates the distribution of bodily fluids, facilitates the transmission of nerve impulses, and ensures both the maintenance of blood pressure and the proper functioning of the cardiovascular system (Harding, 2013).

Beyond its role in nutrition, salt emerges as a crucial component of ancient food practices, particularly in relation to preservation technologies (Alessandri *et al.*, 2019). Salt possesses strong desiccating properties; by extracting water from organic matter, it not only inhibits bacterial growth but also creates conditions that significantly limit the proliferation of microorganisms responsible for food spoilage.

Within many traditional and pre-industrial economic frameworks (Harding, 2021), the preservation of perishable foodstuffs—such as meat, fish, vegetables, and dairy products—therefore emerged as a primary determinant of salt demand. The capacity to extend the durability and storability of food resources represented a critical factor in maintaining food security, enabling the accumulation of surplus production, and fostering the expansion of long-distance exchange and distribution networks.

At the same time, salt played an active role in numerous culinary transformation processes, acting as a fundamental agent in the production of specific foods such as cheese and brined vegetables (Moinier, 2011).

In cheesemaking, the addition of salt to the curd facilitates the separation of whey from the solid mass. Simultaneously, salt forms an external membrane around the curd,

providing immediate protection from external agents, and in those cases where ageing was intended, creating the conditions for the development of a rind (Fox, 1999). In the case of certain vegetables, the canonical example being olives, salt renders otherwise inedible products suitable for consumption by softening their fibres and enhancing both texture and flavour.

The desiccating capacity of sodium chloride was equally essential in non-culinary domains. In tanning, the removal of water from animal hides extended their durability and facilitated further processing. In textile production, salt acted as a catalyst in the chemical interaction between dye and fibre, improving adhesion and ensuring greater resistance to fading (Moinier, 2011).

Archaeological evidence also points to its significance in ceramic manufacture (Erdoğu & Fazlıoğlu, 2006; Stryszewska & Kańka, 2017). Salt was added to the ceramic paste, but in low-temperature ceramics this component was rapidly leached out, leaving behind a porous material. As a result, the vessel surfaces acquired a lighter structure and improved thermal properties in the finished ceramic product.

Finally, salt occupied an important position in medical and ritual practices, where its purifying and preservative properties were symbolically and practically valorised (Dardeniz, 2023; Di Fraia, 2011). Beyond these utilitarian and symbolic dimensions, it also functioned as a key medium of exchange within market systems, acquiring a central role not only as a resource but as a commodity embedded in broader networks of economic and social interaction (Bonghi Jovino, 2002).

Alongside human consumption and use, salt also played a crucial role in animal husbandry. Livestock require sodium supplementation in order to maintain physiological balance, regulate hydration, and sustain growth and milk production, as demonstrated by veterinary studies (Johansson, 2008; Suttle, 2022). For this reason, salt constituted a regular and indispensable component of herd management in both pastoral and mixed farming systems (Marzano, 2013; Eubanks *et al.,* 2025). This function is particularly significant when modelling salt demand, as livestock populations—as will also emerge from the present study—often represent one of the primary consumers of salt within agropastoral economies.

State of the Art in Salt Demand Quantification

Given the importance of salt and its pervasive presence in many aspects of ancient life, it is hardly surprising that scholars have repeatedly attempted to quantify both the production capacity of specific sites and, more broadly, the overall demand for this resource within human communities.

An early and significant attempt at quantification is that of Akridge (2008), who proposed the direct application of a mathematical formula to calculate brine evaporation rates during salt production. Specifically, Akridge (2008) suggested applying to the productive context under examination a formula expressing the amount of salt crystallised (*ms*) as a function of the quantity of water evaporated (*mw*) and the initial concentration of salt in the brine (*S*), which may be summarised as follows:

$$ms = mw(1.52 \times 10 - S^2 + 9.50 \times 10 - 3S)$$

As a case study, Akridge (2008) applied this formula to the site of *La Placita* in Mexico. The production area, consisting of 18 tanks each measuring 3 m, was calculated to yield approximately seven metric tonnes of salt during the dry season of the year 2000 AD. This estimate assumed that 5 days of solar evaporation were necessary to sufficiently concentrate the brine (with an *S* value of ca. 15%), and that each tank would consequently produce 25–30 kg of salt.

However, as B. Moinier (2011) has emphasised, the direct application of such mathematical formulas proves to be exceedingly complex, particularly in cases where salt production involves more intricate technological systems. A notable example is the technique of *briquetage* (Harding, 2021; Weller & Ard, 2024), in which brine evaporation was accelerated not only by solar radiation but also through the boiling of brine in ceramic containers, thereby introducing an additional set of variables related to fuel input and vessel efficiency.

Direct quantification of salt demand therefore presents significant methodological challenges, insofar as it requires the determination of several interdependent factors. Among the most critical factors are (1) the concentration of salt in the available water sources, (2) the efficiency of evaporation processes conditioned by environmental conditions and energy input, (3) the effectiveness of heat application in those cases where boiling was employed, and finally (4) the dimensions and capacities of the containers or evaporation tanks used.

Since precise data on these variables cannot always be retrieved for ancient contexts, most estimates in the literature have been derived indirectly, by reconstructing dietary requirements and evaluating the scale of salt consumption in major craft and industrial activities.

Numerous authors have proposed estimates concerning the amount of salt required for human and animal diets, as well as for specific types of production.

These are summarised in Table 1 and briefly discussed below.

Table 1 Comparison between estimates found in the literature regarding the amount of salt required for its various uses

Reference	Annual salt consumption (kg *pro-capite*)			Salt (kg) used in 1 kg/l production		
	Human	Cattle	Sheep Goat	Bread	Salting meat/fish	Brine
Moinier (2011)	3.2–4.3	-	-	0.09	0.18	2.0
Dardeniz (2023)	3.6	-	-	-	-	-
Ziółkowski (2005)	2.0	30	5	-	-	-
Cifani (2020)	2.5	20	1.4	-	-	-
Di Fraia (2011)	1.1	-	-	-	-	-
Johansson (2008)	-	14.6	7.2	-	-	-
Pasqualone *et al.* (2019)	-	-	-	0.10–0.20	-	-
Alessandri *et al.* (2019)	-	-	-	-	0.30	-
Sternberg (2000)	-	-	-	-	0.40	-

Moinier (2011) advocates the use of information preserved in ancient Greek and Latin texts. According to the French scholar, references to salt in these sources make it possible to calculate the quantities needed both for dietary purposes and for food production within a community, as well as for certain industrial activities.

For example, if the demographic profile of a group is known, this provides what Moinier terms 'a practical calculation basis, mostly concerning the diet', from which one can estimate the salt requirements. This is because the amount 'for human consumption remains rather steady and constant as quantity; the inter-individual variations are not significant in the context of a global approach' (Moinier, 2011).

Drawing on passages from Cato the Elder (*De Agricultura*, LXVII), Aristophanes (*Acharnians*, 719–730), and the *Historia Augusta* (*Life of Claudius Gothicus*, XIV, 1–13), Moinier (2011) estimates that between the last three centuries BCE and the Imperial period (up to the Antonine dynasty), annual per capita salt consumption ranged between 3.2 and 4.3 kg.

This figure is consistent with the estimate of 3.6 kg per year proposed by Dardeniz (2023), but considerably higher than other values reported in the literature. Ziółkowski (2005) and Cifani (2020), for example, suggest lower estimates of 2 kg and 2.5 kg, respectively. The amount of salt consumed annually by a human being must have been around this figure, considering that the recommended daily intake today is 2–3 g per day and Di Fraia's warning (2011) that estimates based on ancient sources are excessively 'inflated', estimating an annual per capita consumption of 1.1 kg. According to Carusi (2018), the values reported in ancient texts do not refer exclusively to food salt, but also include quantities used in other activities.

As regards the salt used in animal husbandry, Moinier (2011) notes that ancient authors and naturalists (*i.e.*, Aristotle, *Historia Animalium* VIII, 10) emphasised its importance, but he is unable to provide precise estimates. By contrast, Ziółkowski (2005) and Cifani (2020) propose annual figures for two species: cattle and sheep/goats. For cattle, the two authors broadly agree, with Ziółkowski suggesting 30 kg per year and Cifani 20 kg. Both values, however, exceed modern veterinary data, such as Johansson's (2008:14) recommendation of 14.6 kg per year. For sheep and goats, the estimates diverge considerably: Cifani proposes a relatively low figure of 1.4 kg per year, while Ziółkowski suggests 5 kg. Once again, both are inconsistent with modern agronomists' guidelines, though in this case they fall short of Johansson's (2008) recommended 7.2 kg per year.

Using ancient sources, Moinier also attempts to quantify the salt required in bread production, which can be reconstructed based on annual grain consumption. Relevant references appear in works such as those of Cato the Elder (*De Agricultura*), Polybius (VI, 39.12–15), and Seneca (*Epistulae Morales ad Lucilium* IX 80.7). Assuming an annual consumption of 420 kg of bread in ancient Italy, Moinier (2011) estimates that around 4 kg of salt would have been required for its production. This corresponds to approximately 9.5 g of salt per kilogramme of bread, a value that aligns with the generally accepted minimum for satisfactory bread-making, 10–20 g per kilogramme (Pasqualone *et al.*, 2019).

More complex is the assessment of salt requirements for the preservation of other foods. Moinier (2011) relies primarily on Cato the Elder (*De Agricultura*) for pork and Columella (*Res Rustica* X, 12.4) for olives in brine. For pork, he estimates a

ratio of 180 g of salt per kilogramme of meat (*i.e.*, 18% of the weight). For olives, the brining solution required 2 kg of salt per litre of water.

In the case of fish preservation, a wide range of salting techniques is documented in antiquity (Carusi, 2008b: 152–53). Fish products could vary considerably, ranging from lightly salted or brined preparations to heavily salted and fully preserved products intended for long-term storage and transport. This variability implies that salt consumption ratios were likely not uniform across all production contexts. Carusi (2008b), for example, drawing on the *Geoponica* (20.46.3), proposes a relatively low ratio of 1:8 (salt to fish), corresponding to approximately 10–11% of the fish's weight. By contrast, Sternberg (2000) suggests a generally applicable ratio of 3:2, indicating that salt could account for approximately 40% of the fish's weight. A slightly lower figure, though of the same order of magnitude, has been proposed by Alessandri *et al.* (2019) for protohistoric Italian contexts, suggesting a minimum salt requirement of about 30% of the fish's weight.

Also, regarding the quantity of salt to be used in leather tanning processes, the ratio between the weight of the product to be treated and the necessary quantity of salt is used. The most precise estimate, although presented in the form of a range, is that of Moinier (2011:142), who proposes a salt percentage equal to 35–45% of the weight of untreated hides.

Methodology: Model Description and Computational Approach

In this study, we opted for an indirect quantification of salt demand to overcome the limitations inherent in a *direct* approach. Rather than relying on ancient literary sources, we adopt a reconstructive model of agropastoral exploitation of the landscape.

This model is designed to estimate:

1. The maximum number of people and animals requiring salt for dietary purposes.
2. The scale of production activities dependent on salt, such as food preparation (*i.e.*, cheese- and bread-making) and preservation techniques (*i.e.*, salting and brining).

It is important to emphasise that the figures presented should be regarded as broad estimates rather than absolute values. In studies of this kind, there is always a margin of variability, or uncertainty, in the results obtained. This is due both to the way input data are collected (*i.e.*, percentages derived from palaeobotanical and archaeozoological datasets) and to the assumptions employed (*i.e.*, annual per capita food requirements, hectares required for pasturing different species, average cereal yields). For this reason, it is essential to indicate the precision of the estimates. In the present study, this margin is quantified at between 5 and 10%. Such a 'standard deviation' must therefore be borne in mind when assessing the reliability of the values proposed and, consequently, of the estimated demand for salt. Nevertheless, the approach provides a general frame of reference against which specific contextual data can be projected, offering insights—at least in terms of trends—into broader historical processes, as well as an indication of the overall scale of salt production.

To obtain all the variables necessary to indirectly quantify a community's salt demand through analysis of its territory, the first step is to generate a model of its agropastoral exploitation.

From a theoretical perspective, the methodology developed and applied in this study is conceptually rooted in the long tradition of palaeoeconomic research that emerged from the mid-twentieth century onward, which placed the organisation and subsistence strategies of past human groups at the centre of scholarly inquiry. Through the spatial assessment of resource availability and exploitable territories, researchers sought to better characterise the organisation of the human communities under study. Foundational works in this regard include those by Clark (1952) and by the Cambridge group of the 'Palaeoeconomy School' (Higgs, 1972, 1975), which established the importance of analysing the relationship between settlements and surrounding productive landscapes to estimate economic sustainability and resource demand.

Similar quantitative and spatial approaches, beginning with the well-known Site Catchment Analysis (Higgs & Vita-Finzi, 1970), were subsequently applied in numerous contexts, increasingly exploring the relationship between settlements and resources through ever more detailed reconstructive models made possible by continuous technological development.

Building upon this research tradition, the model proposed here introduces a further level of methodological development by integrating large datasets and recent GIS-based spatial analysis techniques into a dynamic modelling framework. This approach allows for a more flexible and multiscalar reconstruction of resource demand.

Specifically, this objective is achieved by applying the 'FAO' *land evaluation* techniques (1976), adapted archaeologically (Van Joolen, 2003) according to the methodology developed and illustrated in detail in Sotgia, 2023 and 2024a.

In simplified terms, *land evaluation* (Verheye *et al.*, 2009; Vink, 2013) refers to a technique for establishing the suitability of soils for agropastoral activities, using a system of descending quality classes characterised by one or more limiting factors negatively affecting their use (Cremaschi, 2000), in line with the technical and agricultural knowledge of a given community. The factors considered are both environmental (*i.e.*, soil type, humidity index, altitude, solar exposure) and socio-political (*i.e.*, distance from markets, proximity to road networks).

Through a series of spatial analyses in a GIS framework, a raster image of the terrain is produced, in which each cell (representing a defined spatial unit) is assigned a relative agropastoral suitability score on a scale from 1 (very poor) to 10 (excellent).

In the accompanying supplementary materials, the values employed in the factor analysis, together with the inputs used in the landscape assessment of the case study under consideration, are provided as an illustrative example of the procedures adopted. For a more comprehensive discussion of *land evaluation* and its associated methodologies and procedures, reference should be made to Sotgia, 2024a.

Once this agropastoral model of the study area has been generated, the parameters necessary to quantify the salt demand of the community under study can be calculated through a series of successive steps (Fig. 1).

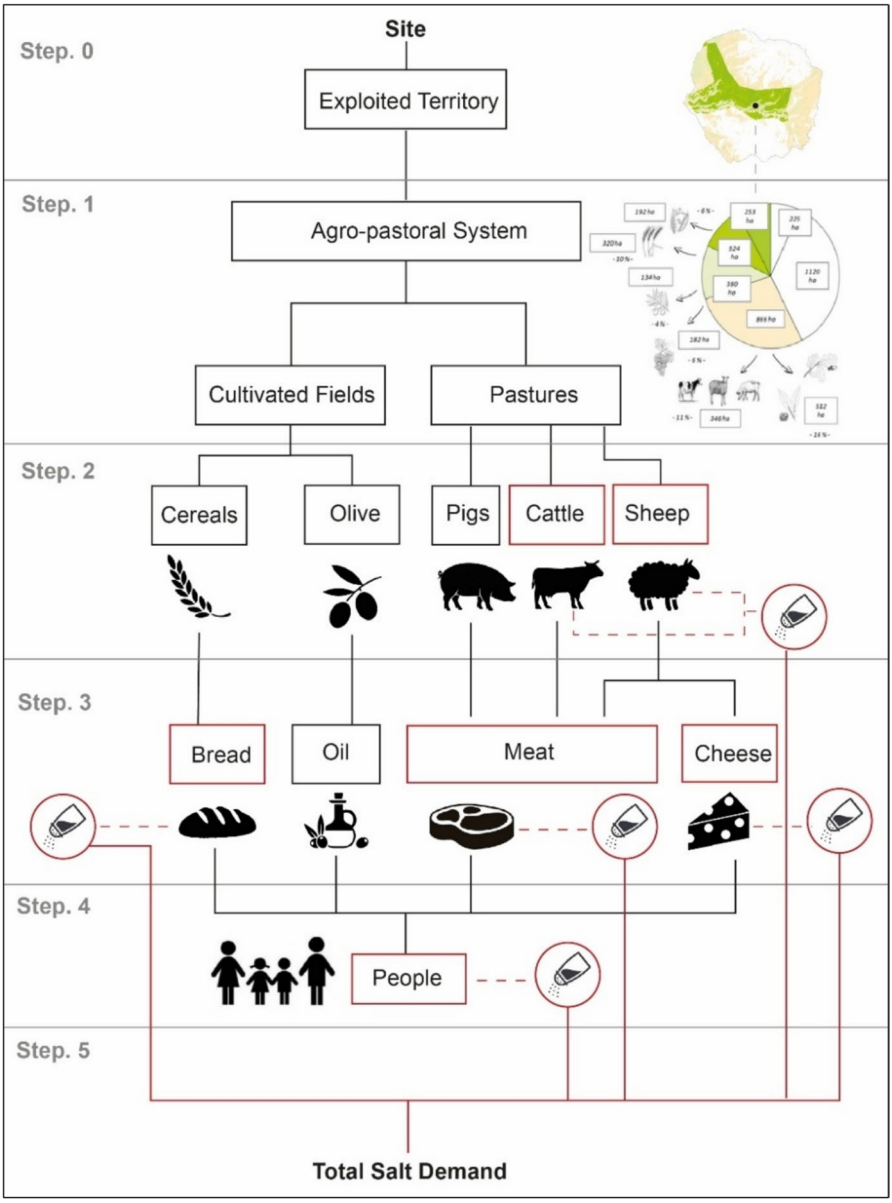

Fig. 1 Infographic showing the steps required to estimate salt demand for the site analysed. The variables that must be considered because they contribute to salt consumption are shown in red

Among the various uses of salt outlined above, the model developed in this study estimates salt demand by focussing exclusively on those applications for which all relevant variables were readily available or could be reliably derived: namely, food-related uses and animal and human consumption. It is important to emphasise,

therefore, that the resulting estimate should be interpreted as a lower-bound approximation of total salt demand. Other uses, such as food preservation and several manufacturing applications, for which the estimation of the relevant variables is not always straightforward, are not incorporated at this stage of the analysis and may account for a substantial additional share of consumption (see 'Discussion' section).

Definition of the Territory Pertinent to a System

The first step is to define the territory exploited by a site or by a system of sites.

Several spatial techniques can be used to calculate the extent of a *site catchment area*, among which the most common are Thiessen/Voronoi polygons and the XTENT model.

For the present model, however, we adopted the *Bubble Method* proposed by Alessandri (2015). Unlike the other techniques—which are based virtually always on a geometric–Euclidean conception of space—this approach allows an analysis that explicitly considers the physical characteristics of the landscape under investigation.

The Bubble Method defines the *area of influence* of a site (*mdi*) based on actual travel distances, primarily on foot, thus considering slopes, obstacles, and specific morphological features rather than relying solely on straight-line (Euclidean) distances. Once the set of sites to be analysed has been defined, the maximum distance that each site exerts over the surrounding territory is determined based on the empirical distribution of average travel times between them.

Accordingly, each site controls a 'bubble' encompassing all land parcels that could be reached within a time interval shorter than the *mdi* corresponding to the period under analysis. For the rules governing the resolution of overlapping bubbles and for a detailed description of the method, see Alessandri (2015).

Extent (in hectares) of the Different Classes of the Agropastoral System

Based on palaeobotanical and environmental data from the study area (*i.e.*, Sotgia, 2024b for the present case study), it is possible to reconstruct the scale of agropastoral activities within the site's catchment area as defined through the Bubble Method.

This involves assigning the cells identified by the model as *suitable for agropastoral exploitation* to specific categories of land use. Only areas with a suitability score ≥ 6 are considered effectively exploited.

The simulation proceeds as follows: a proportion of cells with a value of 6 is allocated to pasture, while the remainder is assumed to be covered by shrubbery (*Macchia Mediterranea*) and forests. Cells with scores of 7 and above are increasingly suitable for cultivation and are divided among crops according to the percentages documented in the palaeobotanical record for different periods.

For the estimation of salt demand, it is particularly important to determine the extent (in hectares) of cereal fields, olive groves, and pastures, since these categories provide the key variables for further analysis. Because each raster cell represents a defined area of space, the total number of cells can be converted into hectares using straightforward mathematical transformations. For example, in our case study, the

spatial resolution of the model is 10 × 10 m; that means each cell represents 100 m^2. Multiplying the number of cells by 0.01 yields the corresponding area in hectares, and thus the extent of the various agropastoral categories.

Total Agropastoral Output: Cereals, Olives, and Livestock

Once the extent of cultivated land has been established, the next step is to quantify the annual agricultural output, defined with reference to one or more yield estimates available in the literature. These estimates are usually expressed as kilogrammes produced per hectare. Knowing the cultivated area therefore allows production to be calculated under different scenarios, depending on which yield estimate is considered most realistic. Numerous studies exist for the Italian context, offering a wide range of comparative values (see Sotgia (2024a:73) for detailed references).

In the present study, rather than adopting a single estimate from the literature, we propose a synthesis based on a combination of previously published data.

For cereal cultivation, we assume that annual output varied according to the land's agricultural suitability: in our model, cells with a value of 7 yield a low output (200 kg), those with a value of 8 a medium output (250 kg), and those with values of 9–10 a high output (300 kg). By dividing the total area of cereal fields among the agricultural suitability classes, it is thus possible to calculate the annual cereal yield (in kilogrammes) for the territory under study.

A similar approach applies to olive production. Once the area of olive groves has been defined, the number of trees per hectare and the average yield per tree can be calculated. Drawing on estimates by Forni (2002) and Keswani (2015), our model assumes a planting density of 40 trees per hectare, each producing 8 kg of olives annually. This gives a yield of 320 kg of olives per hectare under cultivation.

Reconstructing the organisation of animal husbandry requires estimating the total number of livestock sustained by available pasture. Each species is assigned a share of pastureland, based on the hectares required per head, according to the calculations by Cazzella and Moscoloni (1991). The total area of pasture is divided among species in accordance with their representation in the archaeozoological record. From this, the maximum numbers of cattle, sheep/goats, and pigs can be derived, considering species-specific requirements: 1.2 ha per head of cattle, 0.12 ha per sheep/goat, and 0.2 ha per pig (Cazzella & Moscoloni, 1991).

Quantity of Food Produced: Bread, Olives, Meat, and Cheese

Having defined the scale of agricultural production and herd sizes, it is then possible to quantify the annual food output of the territory. For the purposes of estimating salt demand, we focus on those products in which salt was directly used: bread, olives, meat, and cheese.

For cereals, the total annual yield is reduced by one-sixth to account for seed reserved for the following year's sowing and by a further 25% to simulate losses during milling. The resulting figure gives the weight of flour-based products available for human consumption (Sotgia, 2024a).

For olives, it is first necessary to deduct the proportion allocated to oil production. In our simulation, this proportion increases gradually over time to reflect the growing importance of olive oil: from 60% in the earliest period to 80% in the latest.

For meat, different slaughter rates and carcass yields are applied, 50% of cattle at 95 kg per head, 100% of pigs at 23 kg per head, and 30% of sheep/goats at 11 kg per head (Sotgia, 2024a:). Sheep and goats were also exploited for milk, with 30% of the herd assumed to be used for dairying. Each produced around 60 L annually (Minniti, 2012), with cheesemaking requiring 10–15 L of milk per kilogramme of cheese (Fox, 1999). We assume that between 50 and 70% of available milk was transformed into cheese, given the low prevalence of lactose tolerance in Bronze Age adults (Burger *et al.*, 2020).

Maximum Population Sustained

The final parameter required for estimating salt demand is the maximum population that could be supported by the agropastoral system. This is calculated from the annual food output described above and per capita consumption estimates (Sotgia, 2024a).

For cereals, for example, the total available flour is divided by an annual per capita consumption value of 320 kg (based on 800 g daily). To this figure are added the numbers of individuals who could be sustained by meat and dairy products, using conversion factors of 110 g of protein per kilogramme meat, 3.6 g per litre milk, and an annual per capita protein requirement of 18.2 kg.

Although precise estimates of olive consumption are lacking, reliable figures exist for olive oil consumption (Keswani, 2015:13). In our model, we assume an annual per capita consumption of 25 L of oil. Since approximately 7 kg of olives are required to produce 1 L of oil, the total olive yield can be converted accordingly. Summing the contributions from all these food sources yields the maximum population that could theoretically have been sustained annually by the agropastoral system under study.

Total Salt Demand

By employing this model, indirect calculation of the salt demand becomes possible through the estimation of the number of animals and people, as well as the scale of production processes (bread, cheese, and meat) that require salt, according to the productivity estimates presented in Step 2.

In our model, the parameters used predominantly correspond to the mathematical average of all previously published estimates (Table 1). Annual per capita dietary consumption was therefore set at 2.5 kg of salt for humans, 21.5 kg for cattle, and 4.5 kg for sheep and goats. The production of 1 kg bread was assumed to require 13 g salt, while for meat preservation, a ratio of 30% between meat weight and salt was adopted—that is, 300 g salt per kilogramme of meat. Regarding the salt used in cheese production, we considered an intake of around 4%, meaning that each kilogramme of cheese consumed 40 g salt.

With all the necessary variables defined, it is then possible to establish the overall salt demand of a given territorial system.

Once this demand is established, it becomes feasible to estimate the volume of seawater that needs to be processed to obtain the required quantity of salt, bearing in mind that in the Italian context, salt was produced exclusively from marine and brackish water (Bulian *et al.*, 2024), as saltwater springs associated with geological salt deposits are extremely rare or non-existent.

Several estimates exist for the average salt content of 1 m^3 of seawater: Cifani (2020) suggests 30 kg; Akridge (2008) proposes 33–40 kg, while Marzano (2013) gives a much higher figure of 58 kg. In our simulation, however, we adopt a slightly lower value, namely 28 kg of salt per cubic metre of seawater.

This estimate derives from the following reasoning. We start from the specific salinity of Mediterranean seawater, which has been thoroughly studied and is well established in the literature (Lyman, Fleming 1940).

Although precise values vary slightly depending on geographical location, for the purposes of this study—despite the study area being in the western Mediterranean—we primarily rely on compositional data from the eastern Mediterranean (Cotruvo, 2005). This dataset was selected because it represents one of the most detailed and widely cited analyses of Mediterranean seawater chemistry currently available. Moreover, despite regional variability, salinity levels and ionic composition across the Mediterranean basin remain broadly comparable, allowing Eastern Mediterranean values to serve as a reliable proxy for central Tyrrhenian conditions. This seawater contains a mixture of dissolved salts, among which sodium chloride is predominant, followed by magnesium sulphate, magnesium chloride, calcium sulphate, and potassium sulphate, in the proportions reported in Table 2.

We know that 1 mol NaCl consists of 22.99 g sodium (Na) and 35.45 g chlorine (Cl), with a total molecular weight of 58.44 g (Prohaska *et al.*, 2021). However, if Cl is present in excess, not all of it will combine with Na to form NaCl, and the same

Table 2 Composition of seawater (generic and Eastern Mediterranean) with percentage of different elements present

	Typical seawater	Eastern Mediterranean
Chloride (Cl−)	55.04%	54.92%
Sodium (Na+)	30.61%	30.57%
Sulphate (SO42−)	7.68%	7.64%
Magnesium (Mg2+)	3.66%	3.63%
Calcium (Ca2+)	1.16%	1.10%
Potassium (K+)	1.10%	1.20%
Bicarbonate (HCO3−)	0.41%	-
Strontium (Sr2+)	0.04%	-
Bromide (Br−)	0.19%	0.40%
Borate (BO33−)	0.08%	0.19%
Iodide (I−)	-	0.01%

Modified from Cotruvo (2005)

holds when Na exceeds Cl (Pilson, 2012). Calculating the molar ratios for Mediterranean seawater, we find that chlorine is indeed in excess. Theoretically, this would allow for a maximum production of around 30 g of halite per litre (estimated by Cifani, 2020). In practice, however, the actual yield is somewhat lower, since residual brine—containing magnesium and potassium salts (the so-called *bitterns*)—retains also a small quantity of Na and Cl that did not yet precipitate as NaCl but still remains in the brine and is not collected. Under optimal conditions, therefore, the maximum practical yield obtainable from Tyrrhenian seawater is about 28 g NaCl per litre.

Case Study: 'Piscina Torta'

To provide a practical illustration of the model outlined above, we estimated the salt demand of the territorial system associated with the production site of 'Piscina Torta' (*Latium Vetus*, Central Italy—Alessandri *et al.*, 2024) between the second half of the eighth century BC and the second half of the sixth century BC. Initially, the site appears to have been linked to the settlement of Ficana, before being incorporated, following Rome's conquest of Ficana around the mid-seventh century BC, into the broader Roman territorial system (Dionysius of Halicarnassus, *Roman Antiquities* III, 38.3; Livy, *Ab urbe Condita* I, 33.3).

This archaeological site (Fig. 2A), located within the Castel Fusano pinewood on the southern edge of the Tiber delta near Ostia (Rome), is one of the most remarkable protohistoric production areas known along the Tyrrhenian coast of central Italy. Today covered by a dense forest and largely buried underneath dunes inside the Natural Reserve of the 'Roman coast', the site occupies a Holocene beach-ridge system created by successive phases of Tiber delta growth. Geological data indicate that during the late first millennium BCE the shoreline lay considerably farther inland than at present, forming a beach-ridge lagoon system ideal for resource exploitation.

The area first came to scholarly attention in 1984 (Pisani Sartorio& Quilici Gigli, 1984), when surveys revealed 12 dense scatters of ceramic fragments spread over roughly 15 hectares (Fig. 2B). Apart from a few small Roman scatters, the assemblage consisted almost entirely of shards of thick, reddish impasto jars dating from the seventh to the sixth centuries BCE. The overwhelming dominance of storage debris and the absence of domestic structures suggested a specialised activity rather than regular habitation. Pisani Sartorio and Quilici Gigli (1984) then proposed salt production or fish preservation, activities attested in ancient texts and well suited to a lagoonal setting (Alessandri *et al.*, 2019).

Renewed investigations began in 2022 within the 'Salt and Power' project (Alessandri & Attema, 2021), which seeks to understand the network of 'reddish potsherd sites' along the Tyrrhenian coast (Alessandri *et al.*, 2025; Attema *et al.*, 2025; Bulian *et al.*, 2024). At 'Piscina Torta', the team applied a multi-layered methodology: intensive surface collection, drone photogrammetry, geological coring, and high-resolution magnetometry guided by centimetric Real Time Kinematic GPS grids. On the surface, over 14,000 artefacts were systematically recorded, and geophysical prospection revealed rectangular and horseshoe-shaped anomalies, linear

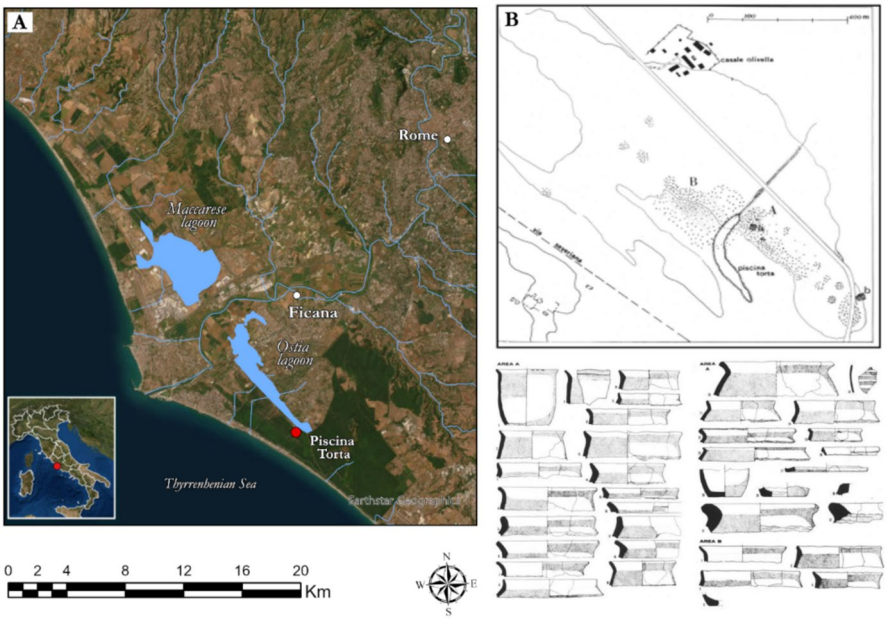

Fig. 2 "Piscina Torta": **A** the locations of the sites mentioned in the text (ESRI satellite images in the background) and **B** the original sketch of the surface materials taken from Pisani Sartorio and Quilici Gigli, (1984) ((a) and (b), Roman materials; (A) and (B), high density of materials)

ditches, pits, and kiln-like structures, all pointing to intensive, spatially organised craft activities.

The recovered vessels, medium-sized *ollae* and large *dolia*, appeared often carefully made, sometimes with notched cord decorations and outcurving rims. The variety of forms differs from the crude, open *briquetage* containers used elsewhere in Europe for salt evaporation and implies that at 'Piscina Torta' a broader range of activities was ongoing. The recovery of pedestals and burnt features consistent with kilns, however, forms solid evidence for salt making, while the fine quality of some of the jars also suits large-scale storage of processed foodstuffs such as fish or other salted products for trading purposes.

Whatever the precise products, 'Piscina Torta' clearly functioned as a specialised production site during a period of rapid urban growth in *Latium*. Sites of this kind likely played a key role in supplying early city-states among which Rome with essential commodities and must have been fundamental for the emerging economies of central Italy. The continuing excavations at 'Piscina Torta' promise to yield crucial insights into the organisation of coastal production and exchange during the Late Iron Age.

To quantify the extent to which this production site could have met regional salt demand, we developed two models of agropastoral exploitation of the surrounding territory, corresponding to the two chronological phases considered: *Period 1* (eighth–seventh centuries BCE), the initial phase of the urban phenomenon—and *Period 2* (seventh–sixth centuries BCE)—the full and mature phase of the

development of the Early States. Each phase was characterised by specific climatic conditions (Bernard *et al.*, 2023) and distinctive modes of resource management (Minniti, 2012; Sotgia, 2024b; Trentacoste, 2020), which in turn affected overall agropastoral yields. Table 3 presents the proportions of animal species and cultivated crops, the organisation of agricultural and pastoral production, and the climatic regime attributed to each period. These parameters provide the 'general rules' for the simulation.

To define the territorial extent of the Ficana and Rome systems, we applied the *Bubble Method*, following the rules and simulation proposed in Alessandri, 2016 and extending the analysis initiated there to cover the entire chronological span considered in this study. By applying the same methodology to the settlements of *Latium Vetus* dated to the seventh and sixth centuries BC—based on the site database published by Palmisano *et al.* (2018)—we estimated that the area controlled by each territorial system extended up to around 2 h of walking distance from the main centre in *Period 1* and up to 3.5 h in *Period 2*.

The boundaries thus calculated for the territorial systems of Ficana and Rome are shown as dashed lines in Figs. 3 and 4. To represent the territorial annexation that followed Rome's conquest of Ficana, which occurred around the mid-seventh century BC, the definition of the Roman territory in *Period 2* was obtained by merging the two bubbles into a single one, according to historical sources (Dionysius of Halicarnassus, *Roman Antiquities* 2.55.5; Livy, *Ab Urbe Condita* 1.33).

During Period 1 (Fig. 3; 750–650 BCE), characterised by a humid climate, the Ficana system extended over 12.596 ha, of which 84% was exploited agropastorally. Within this, 1.206 ha (10%) was woodland, 3,763 ha (30%) pasture, and 5,565 ha (44%) cultivated land. Of the cultivated area, 339 ha was olive groves, 204 ha vineyards, 135 ha orchards, 747 ha legume fields, and 4,140 ha cereal fields. This agropastoral system produced an annual output of 13,000 L olive oil and 575,200 kg bread, and sustained herds of 470 cattle, 1,807 pigs, and 3,577 sheep and goats.

These resources could have supported a maximum population of approximately 2,292 individuals. Considering the settlement area of Ficana (11 ha; Fischer-Hansen & Algreen-Ussing 2013), this figure appears plausible, though slightly higher than the 'standard' estimate (1,100/1,650 inhabitants) derived from the equation proposed by Guidi and di Gennaro (2010), which assumes 100–150 people per defended hectare.

Based on these variables, the hypothetical salt demand of this system is estimated at 56 tonnes annually. Of this, 6 t would have been consumed by humans and 26 t by animals, while 15 t were used in meat preservation, 1 t in cheesemaking, and 8 t

Table 3 Climate conditions and organisation of agricultural and pastoral production in the two different periods analysed

	Climate	*% of animals raised*			*% of cultivated plants*				
		Cattle	*Sheep*	*Pigs*	*Olive*	*Grapes*	*Fruit trees*	*Cereals*	*Legumes*
Period 1: 750–650 BCE	Humid	30	38	29	5	3	2	70	20
Period 2: 650–550 BCE	Dry	13	50	37	10	10	10	60	10

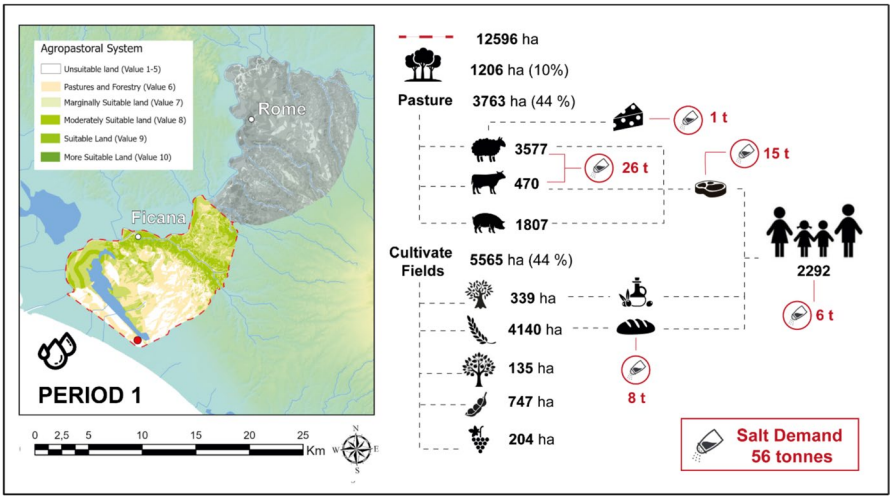

Fig. 3 Period 1: a reconstruction of the agropastoral exploitation of the Ficana area between 750 and 650 BCE and the estimated demand for salt

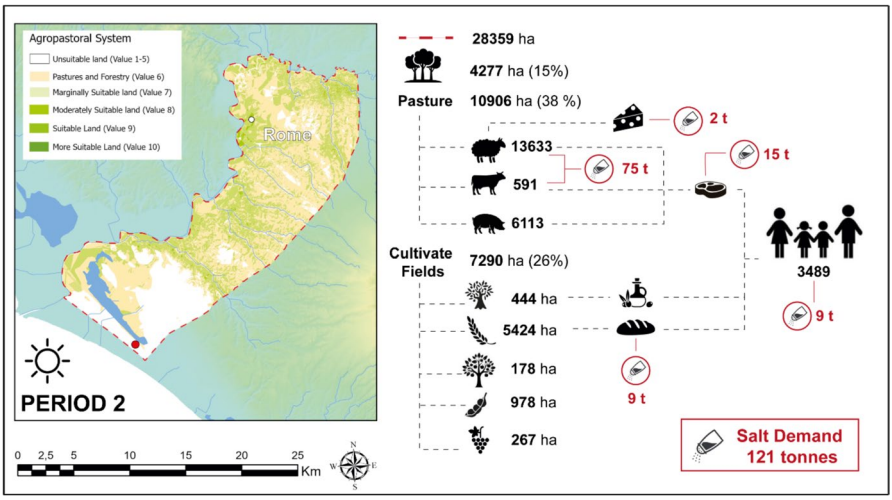

Fig. 4 Period 2: a reconstruction of the agropastoral exploitation of the Ficana area between 650 and 550 BCE and the estimated demand for salt

in bread production. Meeting this demand would have required the processing of at least 2,022 m^3 of seawater.

In Period 2 (Fig. 4; 650–550 BCE), 'Piscina Torta' had become part of the Roman territorial system. As a result of the conquest of Ficana, the landscape controlled by Rome amounts now to 28,359 ha, of which 79% was exploited agropastorally. Within this area, 4,277 ha (15%) was woodland, 10,906 ha (38%)

pasture, and 7,290 ha (26%) cultivation. The cultivated area comprised 444 ha of olive groves, 267 ha vineyards, 178 ha orchards, 978 ha legumes, and 5,424 ha cereals. This arrangement yielded an annual production of 22,758 L olive oil and 682,231 kg bread, and supported 591 cattle, 6,113 pigs, and 13,633 sheep and goats, emphasising the markedly 'pastoral' character of the early Roman community described by Varro (*De Re Rustica* II, 1, 4).

The markedly more pastoral character of landscape exploitation is also due—as noted above—to the effects of a generally drier climate shift. This new climatic set must have resulted in a widespread decrease in the agricultural productivity of the territories. According to our simulation, therefore, this system could have sustained a maximum population of 3,489 individuals. This figure does not always align with demographic estimates derived from the settlement area of Rome in the same period. Without entering into the complexities of the debate surrounding Rome's expansion during its *foundation phases*, it is worth noting the following: our estimate accords with the minimum reconstruction suggested by Ziółkowski (2005), in which only the so-called *civitas Velina* (20 ha) was inhabited, but it is substantially lower than the scenario proposed by Carandini (2012), which envisages the early occupation of the entire *Septimontium* (200 ha).

In any case, the simulated salt demand for this second period amounts to 121 t annually, of which 9 t were required for human consumption and 75 t for livestock. In terms of food production, 26 t were used for meat preservation, 2 t for cheesemaking, and 9 t for bread-making. To satisfy this demand, some 4,253 m^3 of seawater would have needed to be processed, almost double the amount required in the preceding phase.

Before entering the 'Discussion' section, it is worth noting that both estimates of salt demand generated by the model—notwithstanding the limitations that will be discussed below—are broadly consistent in order of magnitude with historical sources. Pliny (*Naturalis Historia* 31, 89), for example, reports that *Ancus Marcius rex salis modios VI in congiario dedit populis et salinas primus instituit* (King Ancus Marcius gave the people six modii of salt as a largess and was the first to establish saltworks).

Given that the *modius*—a Roman unit for dry measures—corresponded to approximately 8.73 L, and assuming a density for salt (using modern coarse kitchen salt as a reference) of around 1.2 kg/l, one *modius* of salt would therefore amount to roughly 10.5 kg. According to Pliny, everyone would have received about 63 kg of salt on this occasion: a substantial grant, presumably sufficient to cover the multiple uses of the resource.

Applying this calculation to the demographic estimates derived in our study, it becomes apparent that, for a first-century writer, a total production in the range of 120 to 220 tonnes would not have been inconceivable. However, the phrase *salinas primus instituit* suggests that, in Pliny's view, such quantities could only be achieved through the method of salt production from saltworks, rather than the *briquetage* technique documented at 'Piscina Torta'.

Discussion

The central aim of this research was to develop an indirect yet archaeologically grounded method for quantifying salt demand through time, based on variables related to agropastoral landscape exploitation. By moving beyond purely qualitative assessments, the proposed model allows demand, production requirements, and patterns of salt use to be expressed in comparable numerical terms, thereby enabling a more precise evaluation of the socio-economic role of salt in protohistoric contexts. As repeatedly highlighted in the literature (see 'Uses of Salt' and 'State of the Art in Salt Demand Quantification' sections), salt was far from being a readily available commodity: it constituted a strategic resource embedded in subsistence practices, craft production, and wider political economies. Quantification is therefore not merely a technical exercise, but a necessary analytical step for understanding the scale, intensity, magnitude, and broader implications of salt exploitation in ancient societies.

Within this framework, the model provides a tool for reassessing wider historical phenomena. For example, the transition from *briquetage* production to saltern-based systems can be interpreted not only as a technological or organisational development within the long-term evolution of salt exploitation—from proto-industrial to fully industrial forms (Olivier, 2025)—but also as a socio-political process. Increasing demand, as reconstructed through the model, implies growing pressures on labour, fuel, and territorial control, helping to frame changes in production strategies as part of broader dynamics of competition, territorial organisation, and the emergence of more complex political and urban entities. These dynamics—as discussed below—are particularly evident in the case studies investigated here: central Tyrrhenian Italy during the first millennium BCE.

A major strength of the proposed methodology lies in its integrative nature. The model brings together heterogeneous datasets—including landscape analysis, paleoenvironmental and productive data, demographic estimates, literary sources, and ethnographic parallels—within a single analytical framework that is relatively straightforward to apply across different contexts.

Although this approach necessarily involves simplifications and schematic representations of complex behaviours, the transparency of the underlying assumptions and the explicit discussion of parameter values (see Supplemental Materials) ensure that the analytical process is fully traceable and replicable. Baseline parameters—such as agricultural yields, per capita consumption, or livestock ratios—can be readily adjusted to explore alternative scenarios, making the model flexible rather than prescriptive.

At the same time, these simplifications also constitute the model's main limitations. In line with broader debates on post-dictive and predictive modelling (see the contributions collected in Citter & Sotgia, 2023), the results are sensitive to the choice of baseline parameters, whose variability can significantly influence quantitative outcomes. Moreover, factors such as climatic fluctuations, extraction technologies, labour organisation, fuel availability, and the socio-commercial dynamics of production cannot always be reconstructed or incorporated into

models with sufficient resolution for ancient contexts. In addition, several uses of salt—such as vegetable preservation, leather tanning, and medicinal or ritual applications—are not fully integrated into the current model and would have further increased overall demand. As a result, the numerical outputs should be interpreted 'only' as orders of magnitude rather than precise values.

Despite these limitations, the quantitative perspective offered by the model provides valuable, albeit approximate, information on the scope of the phenomena analysed. Starting from a basic numerical estimate, rather than simple theoretical assumptions, it will be easier to refine our understanding of the contexts as more detailed data become available, using comparative analytical methods that follow Binford's (2001) 'frame of reference' principle. According to this principle, it is very useful for archaeologists to place their analyses in a broad context and not just keep them limited to specific situations. Projecting data from individual contexts onto a general framework is fundamental if we want to understand when outcomes follow or deviate from more compelling trends.

Moreover, the weaknesses of the model can also be progressively mitigated through continued research and the expansion of available datasets, particularly by integrating experimental and ethnoarchaeological evidence. In this way it will be possible to increase the complexity—and consequent adherence to reality—of models and simulations. Ongoing and future experimental research on salt production, for example, has the potential to refine key variables related to labour investment, fuel consumption, and evaporation efficiency across different production techniques. Through these, the estimation of salt demand can be framed in terms of feasibility and sustainability but also analysed in the more general context of a community's production, estimating its impact on the overall internal organisation.

In this regard, one of the few experimental studies addressing salt production through *briquetage* techniques is the work of Tencariu *et al.* (2015), which provides a useful point of comparison. Although quantitative estimates of *briquetage* production capacity remain scarce, their experiment clearly highlights the high resource intensity of this production method. In their study, only three consolidated salt *briquettes*—each weighing approximately 0.39–0.89 kg—were obtained, despite the prior manufacture of 17 *briquetage* containers. The production of these three blocks required approximately 41 h of manual labour by two individuals, in addition to a considerable consumption of fuel, although the latter was not quantified in detail.

It is therefore evident that *briquetage*-based salt production was extremely resource-intensive, a factor that becomes even more significant when contrasted with the high levels of salt demand, as evident from our case study.

At the same time, comparison with ethnographic or historical cases in which key variables are more tightly controlled offers an important opportunity for cross-validation of the proposed model and its simulations. Such comparisons would allow the model's assumptions and outputs to be tested against well-documented production systems, thereby strengthening its interpretative robustness or, where necessary, prompting revisions to the parameter values employed. In this sense, the approach is deliberately iterative, designed to improve progressively as new and more detailed datasets become available.

Finally, before turning to the discussion of the specific simulation presented here, it is worth emphasising that the proposed methodology can be not limited to salt alone. Because it is grounded in general principles of productive ecology and landscape exploitation, the same analytical logic can be extended to other proto-industrial commodities—such as iron, charcoal, or pottery—whose production depended on quantifiable relationships between environmental inputs, labour, and output. By modelling demand rather than production alone, this framework provides a transferable tool for exploring how increasing resource requirements shaped technological choices, territorial organisation, and socio-political trajectories in ancient economies**.**

All the dynamics outlined above can in general terms be clearly observed in the case study of the production site of 'Piscina Torta'.

First, the quantification of salt demand of the Ficana and Roman territorial systems allows the productive context associated with these systems to be characterised in greater detail. The scale of the production area, the number of identified furnaces, and the extraordinary density of reddish *ollae* fragments have, since the site's discovery, pointed to a high level of technical organisation and productive intensity linked to salt exploitation and to activities connected with marine resources (Pisani Sartorio & Quilici Gigli, 1984; Alessandri *et al.*, 2024). Until now, however, it had not been possible to estimate this production in terms of the actual quantities of salt produced. Through the application of the model, this quantity—albeit in approximate terms—can now be hypothesised, enriching our understanding of the site and allowing for a more complete characterisation of its function. Whether 'Piscina Torta' alone could meet the salt requirements of the Ficana system initially and of Rome subsequently, or whether it operated as part of a wider and still partially unidentified network of production sites remains an open question. Nevertheless, the availability of quantitative estimates provides a new basis for analysing this phenomenon in a more articulated and empirically grounded manner.

An important point in support of the model's validity lies in the consistency between the simulated results and figures derived from ancient sources generally considered reliable (Moinier, 2011). Pliny's account of the distribution of six *modii* of salt as a *congiarium* (*Naturalis Historia* 31, 89), when translated into volumetric and weight estimates and compared with the demographic reconstructions proposed here, suggests quantities that fall within the same order of magnitude as those generated by the model. This convergence offers an independent form of validation and supports the robustness of the proposed methodology.

Second, the gradual yet steady increase in salt demand reconstructed by the model allows us to reflect on salt-production strategies more generally, within the broader technological and productive dynamics of central Italy during the first millennium BCE, and above all on their long-term evolution. The large quantities of salt required in the different periods analysed would have rapidly encouraged the search for more efficient production methods. Beyond the contradictions highlighted by Olivier (2025), related to seasonality and labour organisation, *briquetage* was further constrained by its heavy fuel consumption in relation to the amount of salt produced, as also suggested by the data from the experiment by Tencariu *et al.* (2015). Large quantities of wood were required to accelerate brine evaporation, and

although precise quantitative assessments of fuel use are still lacking, it is reasonable to assume that beyond a certain threshold this practice became environmentally and economically unsustainable. This limitation would have been particularly acute in contexts where wood was simultaneously required for metallurgy, construction, and other productive activities.

If the volumes of seawater implied by the simulations are accepted—volumes that reach magnitudes at which *briquetage* becomes increasingly inefficient—it becomes necessary to posit a transition towards saltern-based production.

In this respect, the context of 'Piscina Torta' can once again be used as a key example. The proximity of brackish wetlands around the site may have favoured their use as proto-*salinae*, representing an intermediate technological and organisational stage between *briquetage* and fully developed saltern systems. It is therefore plausible that the transition to this new mode of production should be placed within the Archaic period. Rather than viewing this transition as abrupt, it is more likely to have been a gradual process of experimentation and adaptation, in which sites such as 'Piscina Torta' functioned as laboratories for developing increasingly efficient solutions to the growing demands of more complex societies. In this perspective, it is also possible that in the later phases *briquetage* continued to be employed on a limited scale, perhaps for the production of finer-quality salt intended primarily for human consumption.

More generally, the patterns observed at 'Piscina Torta' mirror a broader trend evident across central Tyrrhenian Italy during the Late Iron Age and Archaic periods. The scale of salt demand reconstructed for the phases analysed—one corresponding to the formative stages of urban communities (*Period 1*) and the other to the full maturation of this process (*Period 2*)—is substantial and allows the hypothesis that control over salt production played a decisive role in territorial organisation and in the development of early state communities to be supported by quantitative data (see also Attema *et al.*, 2025). The role of salt in socio-political dynamics is visible both in settlement patterns—such as the proliferation of 'secondary' coastal centres from the ninth century BCE onwards—and in interurban relations, including Rome's aggressive expansion towards the coast at the expense of neighbouring systems such as Ficana and Veii. In this context, salt emerges not merely as a subsistence resource, but as a strategic commodity whose production and distribution were deeply embedded in processes of political centralisation and territorial competition.

The case of 'Piscina Torta' thus demonstrates the potential of quantified production and consumption data for interpreting settlement dynamics through the lens of resource control. At the same time, it is crucial to reiterate that salt represents only one component of a much wider economic system. Its analytical value lies precisely in this fact. At 'Piscina Torta' and at comparable sites, salt can be understood as a proxy for food preservation within economies geared towards surplus production. The quantities involved imply not only direct consumption, but also large-scale practices of transformation, storage, and redistribution, aimed at supplying both expanding urban centres and their rural hinterlands. In this sense, salt production is inseparable from broader systems of provisioning and surplus management that underpinned early urban growth.

Extending this approach to other commodities and regions would therefore allow the economic foundations of geopolitical change—at the critical transition between protohistory and the Roman period—to be explored with greater analytical precision. At the same time, the application of the model highlights a significant missing link between production and consumption: trade and circulation. While the model estimates how much salt was needed and potentially produced, it does not yet explicitly account for the mechanisms through which salt was transported, exchanged, and controlled across space. Given the strategic value of salt, trade routes—whether coastal, fluvial, or terrestrial—must have played a fundamental role in shaping systems of resource control. Integrating modelling approaches that address mobility, transport costs, and exchange networks therefore represents a crucial avenue for future research.

Conclusion

This study has proposed and tested an indirect, landscape-based methodology for quantifying ancient salt demand, demonstrating its potential to bridge archaeological data, environmental reconstruction, and socio-economic interpretation. By modelling agropastoral production and translating it into estimates of consumption and productive requirements, the approach moves beyond purely qualitative assessments and provides a replicable framework for evaluating the scale and intensity of salt exploitation in protohistoric contexts.

Applied to the case of Piscina Torta, the model highlights how relatively modest changes in territorial extent, climate, and demographic scale could generate disproportionate increases in demand for a strategic resource. These results offer new insights into the economic pressures underlying settlement reorganisation, territorial expansion, and the emergence of early urban systems in central Tyrrhenian Italy.

In particular, they support the interpretation of salt as a key driver in the political economy of early Rome, closely linked to processes of conquest, integration, and resource control.

More broadly, the study underscores the value of salt as an analytical proxy for surplus-oriented economies, in which food preservation, storage, and redistribution played a central role. While salt is only one commodity among many, its pervasive use across dietary, productive, and exchange systems makes it especially well-suited for quantitative analysis. At the same time, the research identifies clear avenues for future development, notably the integration of trade networks and circulation dynamics, which remain essential for fully understanding the relationship between production, consumption, and power.

Ultimately, the methodology presented here is not confined to salt alone. Its principles can be extended to other proto-industrial commodities whose production depended on measurable relationships between environment, labour, and output. In this sense, the model offers a transferable analytical tool for exploring how increasing resource demands shaped technological choices, territorial organisation, and geopolitical trajectories at the threshold between protohistory and the Roman world.

Supplementary Information The online version contains supplementary material available at https://doi.org/10.1007/s10816-026-09771-7.

Author Contribution A.S.: conceptualization, methodology, validation, formal analysis, writing—orginal draft, data curation, and visualisation; L.A: writing—review and editing, data curation, and funding acquisition; P.A.: writing—review and editing, project administration, and funding acquisition; F.B.: writing—review and editing and data curation; J. S.: writing—review and editing and data curation

Funding This paper was written as part of the research scheme 'Salt and Power, Early States, Rome and Resource Control', which was fully funded under the Open Competition Scheme (file nr. 406.20.HW.013) of the Dutch Research Council (NWO) during the period 2021–2026.

Data Availability The authors confirm that all data generated and analysed, as well as the methodologies necessary to replicate the study, are included in the supplementary materials.

Declarations

Competing Interests The authors declare no competing interests.

Open Access This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by/4.0/.

References

Akridge, D. G. (2008). Methods for calculating brine evaporation rates during salt production. *Journal of Archaeological Science, 35*, 1453–1462.

Alessandri, A. (2015). Exploring territories: Bubble model and minimum number of contemporary settlements: A case study from Etruria and Latium Vetus from the Early Bronze Age to the Early Iron Age. *Origini, XXXVII*, 175–199.

Alessandri, L. (2016). Hierarchical and federative polities in protohistoric Latium Vetus. An analysis of bronze age and early iron age settlement organization, in Attema P., Seubers J, Willemsen S. (eds.) *Early states*, *territories and settlements in protohistoric Central Italy. Proceedings of a specialist conference at the Groningen Institute of Archaeology of the University of Groningen*, *2013*, Barkhuis, Groningen.

Alessandri, L., & Attema, P. (2021). Dal briquetage alle saline nella protostoria dell'Italia centrale. *Ipo-TESI di Preistoria, 14*(1), 161–168.

Alessandri, L., Achino, K. F., Attema, P. A. J., Nascimento, M., Gatta, M., Rolfo, M. F., Sevink, J., Sottili, G., & Gorp, W. (2019). Salt or fish (or salted fish)? The Bronze Age specialised sites along the Tyrrhenian coast of Central Italy: New insights from Caprolace settlement. *PLoS One, 14*, Article e0224435.

Alessandri, L., Attema, P., Bulian, F., Sevink, J., De Neef, W., Baiocchi, V., Rolfo, M. F., Cifani, G., Ceccato, Z. L. A., Cusimano, L., De Vos, M., Di Giacomo, L., Fiorillo, A., Gianni, V., Improta, C., Rossi, C., Ter Horst, Y., & Vagliviello, S. (2024). Salt in Late Iron Age Italy. A multidisciplinary approach to the exploration of Italy's coastal exploitation sites: 'Piscina Torta' (Ostia, Rome) case study. *Journal of Archaeological Science: Reports, 53*, Article 104361.

Alessandri, L., Bulian, F., Attema, P., & Sevink, J. (2025). Salt production, trade and consumption in European pre-industrial societies: Analytical methods and socioeconomic contexts. *Quaternary Science Reviews, 356*, Article 109291.

Attema, P., Alessandri, L., Bulian, F., Sevink, J., & Sotgia, A. (2025). Production and demand of salt in ancient Italy from the bronze age to the Roman period. in Eubanks *et al.* 2025, 211–233

Belardelli, C., Alessandri, L., Aranguren, B., Attema, P., Cinquegrana, M. R., Montagnari Kokelj, M., Negroni Catacchio, M., Pacciarelli, M. (2021). Il Sale. Record archeologico, produzione e manipolazione, in I. Damiani, A. Cazzella, V. Copat (Eds.), *Preistoria del cibo: L'alimentazione nella preistoria e nella protostoria* (Studi di Preistoria e Protostoria 6), Istituto Italiano di Preistoria e Protostoria, Firenze, 573–584.

Bernard, S., McConnell, J., Di Rita, F., Michelangeli, M. D., Sadori, L., Masi, A., Zanchetta, G., Bini, M., Celant, A., Trentacoste, A., Lodwick, L., Samuels, J. T., Mariotti, L. M., Bellini, C., Paparella, C., Padilla, P. D., Tan, J., van Dommelen, P., De Giorgi, A. U., & Cheung, C. (2023). An environmental and climate history of the Roman expansion in Italy. *Journal of Interdisciplinary History, 54*(1), 1–41.

Binford, L. (2001). *Constructing frames of references. An analytical method for archaeological theory building using ethnographic and environmental data sets*. University of California Press.

Bonghi Jovino, M. (2002). Tarquinia, sale e saline. In P. G. Michelotto (Ed.), *Λόγιος ἀνήρ. Studi di antichità in memorie di Mario Attilio Levi, Cisalpino,* 27–37.

Bulian, F., Alessandri, L., Attema, P., & Sevink, J. (2024). Bronze age to Roman period salt production in the coastal areas of peninsular Italy: Palaeoenvironments, production methods and archaeological evidence. *Quaternary Science Reviews, 344*, Article 108930.

Burger, J., Link, V., Blocher, J., Schulz, A., Sell, C., Pochon, Z., Diekmann, Y., Žegarac, A., Hofmanová, Z., Winkelbach, L., Reyna-Blanco, S., Bieker, V., Orschiedt, J., Brinker, U., Scheu, A., Leuenberger, C., S. Berino, T., Bollongino, R., Liedke, G., … Wegmann, D. (2020). Low prevalence of lactase persistence in Bronze Age Europe indicates ongoing strong selection over the last 3,000 years. *Current Biology, 30.21*, 4307-4315.e13.

Carandini, A. (2012). *Atlante di Roma antica*. Elceta.

Carusi, C. (2008). Intorno alla produzione di sale a Populonia e nell'Ager Cosanus: Due casi di studio a confronto. In V. Acconcia & C. Rizzitelli (Eds.), *Materiali per Populonia 7* ,Edizioni ITS, 7–16

Carusi, C. (2008b). *Il sale nelmondogreco (VI a.C. – III d.C.). Luoghi di produzione, circolazionecommerciale, regimi di sfruttamentonelcontestodelMediterraneo*. Edipuglia.

Carusi, C. (2018). Salt and fish processing in the ancient Mediterranean: A brief survey. *Journal of Maritime Archaeology, 13*, 481–490.

Cazzella A., Moscoloni M. (1991), Aspetti dell'economia di sussistenza durante l'età del Bronzo a Coppa Nevigata e nell'Italia meridionale, *Scienze dell'Antichità 5*, 233–264.

Cifani, G. (2020). *The origins of the Roman economy: From the Iron Age to the Early Republic in a Mediterranean perspective*. Cambridge University Press.

Cinquegrana, M. R. (2025). *Siti costieri dell'Italia tirrenica dal Bronzo finale all'età del Ferro. Casi studio dall'Etruria settentrionale (Golfo di Follonica).* Phd Thesis, University of Groningen, Groningen. https://doi.org/10.33612/diss.1310548970

Sotgia, A., & Citter, C (2023). *Modelling the landscape. From prediction to postdiction. In: Citter C., Sotgia A. (eds.), Proceedings of the international session at 7th landscape archaeology conference (Iași, 10–15 September 2022)*, Archeologia e Calcolatori 34.1, All'Insegna del Giglio.

Clarke, J. G. D. (1952). *Prehistoric europe. The economic basis*. Methuen.

Cotruvo J. A. (2005). Water desalination processes and associated health and environmental issues, *Water conditioning & purification international magazine* – January.

Cremaschi, M. (2000). *Manuale di geoarcheologia*. Laterza.

Dardeniz, G. (2023). A model of salt production and consumption patterns in Bronze Age Anatolia. *European Journal of Archaeology, 26*, 39–56. https://doi.org/10.1017/eaa.2022.27

Di Fraia, T. (2011). Salt production and consumption in prehistory: Towards a complex systems view. *Exotica in prehistoric Mediterranean* (pp. 26–32). Oxbow Books.

Erdoğu, B., & Fazlıoğlu, I. (2006). The Central Anatolian salt project: A preliminary report on the 2004 and 2005 surveys. *Anatolia Antiqua, 14*, 189–203.

Eubanks, P. N., Dumas A. A., McKillop H., Alexianu M. (eds.) (2025). Meridians *of salt: Global perspectives on archaeology and ethnoarchaeology*, Springer Nature, Switzerland.

Fischer-Hansen T., Algreen-Ussin G. (eds.) (2013). *Excavations at Ficana*, *vol. 3: The iron age fortifications*, Edizioni Quasar

Forni G. (2002). Colture, lavori, tecniche, rendimenti. in Forni G., Marcone A., (eds.) *Storia dell'agricoltura italiana: l'età antica*, 63–148.

Fox, P. F. (1999). Major cheese gropus. *Cheese: Chemistry, physics and microbiology.* (Vol. Volume 2). Aspen Publishers.

Guidi, A., & di Gennaro, F. (2010). Lo stato delle anime come mezzo per la ricostruzione della popolazione dei villaggi preistorici. *Arqueología Espacial, 28*, 351–358.

Harding, A. (2013). *Salt in prehistoric Europe*. Sidestone Press.

Harding, A. (2021). *Salt: White gold in early Europe*. Cambridge University Press.

Higgs, E. S. (1972). *Paper in economic prehistory*. Cambridge University Press.

Higgs, E. S. (1975). *Paleoeconomy*. Cambridge University Press.

Higgs, E. S., & Vita-Finzi, C. (1970). Prehistoric economy in the Mt. Carmel area of Palestine: Site catchment analysis. *Proceedings of the Prehistoric Society, 36*, 1–37.

Johansson, K. (2008). *Salt to ruminants and horses*. Department of Anim. Physiology. Swedish Univ. Agric. Sci.

Keswani, P. (2015). Olive production, storage, and political economy at Late Bronze Age Kalavasos, Cyprus. In A. Jacobs & P. Cosyns (Eds.), *Cypriot material culture studies from picrolite carving to proskynitaria analysiS* (pp. 1–24). Brussels University Press.

Lyman, J., & Fleming, R. H. (1940). Composition of sea water. *Journal of Marine Research, 3*(2), 134–146.

Mandolesi, A. (2024). *Tra Caere e Tarquinia. La costiera civitavecchiese in età etrusca*. All'Insegna del Giglio.

Marzano, A. (2013). *Harvesting the sea: The exploitation of marine resources in the Roman Mediterranean*. Oxford University Press.

Minniti, C. (2012). *Ambiente, sussistenza e articolazionesocialenell'ItaliacentraletraBronzomedio e Primo Ferro*. Bar Publishing.

Moinier, B. (2011). Salt in the antiquity: A quantification essay. In M. Alexianu, O. Weller, & R. Curcă (Eds.), *Archaeology and anthropology of salt: A diachronic approach* (pp. 137–147). BAR Publishing.

Olivier, L. (2025). The Briquetage de la Seille (Lorraine, Eastern France). in *Eubanks et al.* 2025, 235–256.

Pacciarelli, M. (1991). Territorio, insediamento, comunità in Etruria meridionale agli esordi del processo di urbanizzazione. *Scienze dell'Antichità, 5*, 163–208.

Palmisano, A., Bevan, A., & Shennanet, S. (2018). Regional demographic trends and settlement patterns in Central Italy: Archaeological sites and radiocarbon dates. *Journal of Open Archaeology Data, 6*, Article 2.

Pasqualone, A., Caponio, F., Pagani, M. A., Summo, C., & Paradiso, V. M. (2019). Effect of salt reduction on quality and acceptability of durum wheat bread. *Food Chemistry, 289*, 575–581.

Pilson, M. E. (2012). *An introduction to the chemistry of the sea*. Cambridge University Press.

PisaniSartorio, G., & Quilici Gigli, S. (1984). Trovamenti arcaici nel territorio laurentino: Annotazioni di topografia e prospettive di ricerca. *Bullettino della Commissione Archeologica Comunale di Roma, 89*, 9–26.

Prohaska, T., Irrgeher, J., Benefield, J., Böhlke, J. K., Chesson, L. A., Coplen, T. B., Ding, T., Dunn, P. J., Gröning, M., Holden, N. E., & Meijer, H. A. (2022). Standard atomic weights of the elements 2021 (IUPAC technical report). *Pure and Applied Chemistry, 94*(5), 573–600.

Sotgia, A. (2023). A predictive model to investigate the agro-pastoral exploitation of ancient landscapes. *Citter, Sotgia, 2023*, 267–278.

Sotgia, A. (2024a). *The agro-pastoral exploitation of pre-Etruscan Southern Etruria: GIS land evaluation models for the final bronze and early iron ages*. BAR Publishing.

Sotgia, A. (2024b). Notes on agriculture in Etruria and Latium Vetus between the Late Bronze and Iron Age (11th - 8th centuries BC). *Tijdschrift Voor Mediterrane Archeologie Supplement, 3*, 99–106.

Stenberg, M. (2000). Données sur les produits fabriqués dans une officine de Neapolis (Nabuel, Tunisie). *Mélanges De L'école Française De Rome - Antiquité, 112*(1), 135–153.

Stryszewska, T., & Kańka, S. (2017). The effects of salt crystallization in ceramic bricks in terms of line deformations. *Procedia Engineering, 193*, 120–127.

Suttle, N. F. (2022). *Mineral nutrition of livestock* (5th edition). CAB International.

Tencariu, F., Alexianu, M., Cotiugă, V., Vasilache, V., & Sandu, I. (2015). Briquetage and salt cakes: An experimental approach of a prehistoric technique. *Journal of Archaeological Science, 59*, 118–131.

Trentacoste, A. (2020). Fodder for change: Animals, urbanisation, and socio-economic transformation in protohistoric Italy, .*Theoretical Roman Archaeology Journal, 3*(1), 1, 1–17.

Van Joolen E. (2003). *Archaeological land evaluation: A reconstruction of the suitability of ancient landscapes for various land uses in Italy focused on the first millennium BC*, Phd Thesis, University of Groningen, Groningen. https://research.rug.nl/en/publications/archaeological-land-evaluation-a-reconstruction-of-the-suitabilit/

Verheye, W., Koohafkan, A. P., & Nachtergaele, F. (2009). The FAO guidelines for land evaluation. *Encyclopedia of Land Use, Land Cover and Soil Sciences: Land Evaluation, 2*, 78–100.

Vink, A. P. (2013). *Land use in advancing agriculture* (Vol. Vol. 1). Springer Science & Business Media.

Weller, O., & Ard, V. (2024). Prehistoric salt production: Technological approach in ceramic studies. *Journal of Archaeological Science: Reports, 53*, Article 104379.

Ziółkowski, A. (2005). *Historia Rzymu*. Poznań.

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.